Nicholas Chin Bo You

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Education

University of Utah — *August 2024 – Present*
Salt Lake City, UT

BS in Physics with Computational Physics and Astronomy Emphasis
BS in Applied Mathematics

- CGPA: 3.896
- Dean's List Fall 2024, Spring 2025, Fall 2025, Spring 2026

Foon Yew High School — *Jan 2018 - Nov 2023*
Johor Bahru, Malaysia

Selected Research & Publications

Secondary Analysis of Extended Source GEMINGA at VERITAS (SURF Program)
Investigating extended gamma-ray emission associated with the Geminga pulsar using very-high-energy gamma-ray observations.

Searching for Photospheric Pulsation Modes of β Canis Majoris with Stellar Intensity Interferometry
Refining SII data-analysis workflows for the β Cephei variable β Canis Majoris to extract modulation signatures associated with photospheric pulsations.

Efficient Quantum State Preparation for Astronomical Image Encoding
Studying sparse state-preparation methods for image encoding to reduce circuit complexity by exploiting the nonzero structure of sparse images.

Shared Risk-Capital Constraints in Dynamic Option Dealer Hedging
Formulated dynamic option-dealer hedging with shared funding limits as a shared-constraint generalized Nash equilibrium problem and analyzed its variational equilibrium characterization.

Measurement of the photosphere oblateness of γ Cassiopeiae via Stellar Intensity Interferometry with the VERITAS Observatory *A. Archer et al 2025 ApJ 995 191*
*A. Archer, J.P. Aufdenberg, . . . , **N. B. Y. Chin**, . . . , The VERITAS Collaboration*

Presentations

Quantum Computing Focused Reading Group — Spring 2026

- Affliated with NSF Research Training Group

APS Four Corners Section 2025 Annual Meeting — Fall 2025
Poster Presentation

University of Utah Department of Physics and Astronomy Annual Symposium — Spring 2025

Skills

- Languages: English, Mandarin Chinese, Cantonese Chinese, Hakka Chinese, Bahasa Melayu
- Programming Languages: Python, C/C++, Rust, Java, Bash
- Web & Markup: LaTeX, HTML, CSS
- Scientific Computing : CERN ROOT, Gammapy, Jupyter Notebook, NumPy, Matplotlib, MATLAB
- Quantum Computing: Qiskit
- Operating Systems: Linux / Unix

Certificates and Achievements

Summer Undergraduate Research Fellowship (SURF)	Summer 2026
Thomas J. Parmley Scholarship Endowment Fund	Fall 2026 - Spring 2027
MIT iQuHACK 2026 - BlueQubit Problem	World Rank 35, 2nd Highest Score
Undergraduate Research Opportunity Program Award (UROP)	Spring 2026
Talent Scholarship	Fall 2025 - Spring 2026

Chen Jingrun's Cup Secondary School Mathematics Competition 2021	National Rank 81st
Chen Jingrun's Cup Secondary School Mathematics Competition 2023	High Distinction
OpenAI Codex Hackathon	1st runner up
IELTS 2024	6.5
MATLAB Onramp	Course Completion Certiicate

Experience

Research Assistant — *Salt Lake City, UT*
Measuring Stellar Sizes using ACT arrays as Intensity Interferometers — *Nov 2024 – Present*

- Working with Professor LeBohec and Professor Kieda on refining data analysis process using CERN ROOT and C++.
- Developed C++/ROOT library to streamline analysis processes.
- Implemented an FPGA-to-ROOT file conversion tool.
- Measuring photosphere modulations of variable stars.

Undergraduate Student Advisory Committee - Chair — *U of U*
U of U Physics Department — *Oct 2025 – Present*

- Expanded committee membership by 300% through active recruitment and outreach initiatives.
- Organized a department-wide social event in collaboration with the Society of Physics Students (SPS), attracting 30+ participants across undergraduate, graduate, and non-physics programs.
- Led a student town hall attended by 40+ students to collect and communicate departmental feedback to faculty leadership.

Blockchain Club - Officer — *U of U*
Fintech Club - Treasurer — *Jan 2025 – Jun 2025*
U of U Finance Department — *Jul 2025 – Present*

- Developed educational content introducing blockchain, frontier science, and emerging computational technologies.
- Authored educational articles for the club's official website.

UCubesat Club - CFO — *U of U*
U of U ECE Department — *Jan 2025 – Present*

- Telemetry Team; Ground Station Subsystem Team Lead.
- Working on building a ground station to contribute to the SatNOGS Network.

Engineering Possibilities Summit Scholar — *Jan 2025 - Dec 2025*
Goldman Sachs

Apprentice Management — *Johor Bahru, Malaysia*
Rhapsody Youth Ensemble — *2 projects (Approximately 6 months)*

- Established the foundational framework for ensemble's operations and concert planning.

Co-Chairman and President of Education — *Johor Bahru, Malaysia*
Foon Yew Symphony Orchestra — *June 2022 - May 2023*

- Utilized recording and mixing skills to support the orchestra during the COVID period, facilitating virtual performances
- Organized and conducted more than 10 hours of online courses on different subjects weekly to maintain member engagement and skill development.
- Designed programs to help students foster interest and performing opportunities. Approximately 80% of members increased 40% more performing opportunities annually

Assistant Luthier and Sales Assistant — *Johor Bahru, Malaysia*
Melodia String — *Jan 2024 - Aug 2024*

- Performed walk-in instrument maintenance and basic repairs, including string replacement, setup adjustments, and minor restorations
- Studied and completed projects up to an intermediate level luthiery techniques
- Maintained workshop equipments and tools

Recording Studio Assistant — *Johor Bahru, Malaysia*
Goldstone Music — *Jan 2024 - Aug 2024*

- Assisted with recording session setup, including microphone placement, signal routing, and basic audio equipment configuration

- Provided technical and logistical support during live stage performances and events
- Studied music arrangement and audio recording techniques through hands-on observation and guided practice

Projects

Option Pricing using Quantum Computers *Option-Pricing-on-Quantum-Computer*

- A from-scratch implementation of Stamatopoulos et al. (2020). "Option Pricing using Quantum Computers"

A Guide to Quantum Computing *A-Guide-to-Quantum-Computing*

- A collection of notes and implementations aimed at building intuition for quantum computing concepts without requiring prior knowledge of quantum mechanics.

A Complete DNS Server *DNS-Server-in-RUST*

- Developed a fully functioning DNS Server that handles DNS queries, resolves domain names, and returns IP addresses following standard DNS protocol specifications with Rust.

Relevant Coursework

Foundations of Analysis I	In Progress
Intro to Probabilities	In Progress
Intro to Quantum Computers - Graduate Course	A
Enhanced Intro to Differential Equations - Honors Course	B
Modern Physics	A
Survey of Numerical Analysis	A
Enhanced Linear Algebra - Honors Course	B+
Foundations of Astronomy	A
Accelerated Object-Oriented Programming	A-
Physics I, II	A
Physics I Lab	A
Calculus I, II, III	A